

A NEW CATEGORY BETWEEN EXTERNAL FIXATION AND DEFINITIVE RECONSTRUCTION

Temporary Internal Stabilisation for Critical Long-Bone Defects

A modular intramedullary system designed for staged reconstruction, PMMA spacer support, and infection-conscious temporary fixation.



TRL 4-5

Validated

PMMA Ready

Modular

HA-Ag Coating

TREATMENT PATHWAY

External Fixation
Current standard



CITadel (TIS)
Temporary internal stabilisation



Definitive Reconstruction
Bone transport / graft

KEY FEATURES



Modular Architecture

Two-component system adaptable to defect size and anatomy.



Mechanical-Axis Support

Intramedullary alignment for stable temporary fixation.



PMMA Spacer Compatibility

Designed to function with antibiotic-loaded spacers.



HA-Ag Antibacterial Interface

Selective coating in bone-contact zones.

DIFFERENTIATION

✗ Current Limitations

- ✗ External fixation dependency
- ✗ Lack of spacer-compatible systems
- ✗ Limited internal options for defects

✓ CITadel Advantages

- ✓ Internal temporary stabilisation
- ✓ Designed for spacer-based workflows
- ✓ Defect-focused construct logic

CLINICAL USE CASES



Infected Critical Bone Defects

Osteomyelitis cases requiring staged debridement, antibiotic spacer support, and delayed reconstruction.



Blast & High-Energy Trauma

Complex limb injuries with segmental bone loss requiring temporary internal stabilisation.



Oncologic Segmental Resection

Post-resection defects needing stable internal fixation during staged limb salvage.

CLINICAL & ECONOMIC IMPACT



Reduced hospital stay



Faster patient recovery



Improved stability



Fewer complications

VALIDATION STATUS

TRL 4-5

Bench Testing Completed

Early Clinical Application

Prototype Validated

CITadel has undergone biomechanical testing and early clinical use.

Current TRL: 4-5 → Target: TRL 6 (clinical pilot)

PARTNERSHIP OPPORTUNITIES



Industrial Licensing

License for manufacturing and distribution within your product portfolio.



Co-Development

Joint engineering and regulatory pathway towards CE/MDR compliance.



Clinical Pilot Sites

Reference centre for structured clinical validation and pilot deployment.



Strategic Investment

Accelerate transition to clinical-scale adoption through strategic funding.



Maksym Baida, PhD

ORTHOPAEDIC TRAUMA SURGEON • DEVELOPER OF CITADEL

Over 15 years of clinical practice in traumatology and reconstructive orthopaedics. Associate Professor at Bogomolets National Medical University, Kyiv. CITadel was born from a direct clinical need – the absence of a reliable internal stabilisation option for patients requiring staged reconstruction.

Let's Start the Conversation

CITadel is open for collaboration with industry partners, clinical centres, and strategic stakeholders.

